

Doom PWAD QA Report

Run 33675e79-9272-4e02-bedd-d51e89d1449f | Map E1M1 | Outcome timeout

Executive Summary

This is an autonomous QA test of E1M1 using an LLM-directed test harness. The test was designed to evaluate map navigation, combat engagement, resource balance, and secret discovery, resulting in a timeout outcome after 22 seconds.

Pass/fail:

map_navigation: PASS

combat_engagement: FAIL

resource_balance: FAIL

secret_coverage: PASS

overall_verdict: PARTIAL

- The automated playthrough moved to 4 distinct position clusters, satisfying the navigation pass criteria despite the timeout.
- Only 1 of 8 enemies were killed, resulting in a 12.5% engagement rate which is below the 50% threshold.
- The health and ammo ratios were 0.0, failing to meet the minimum threshold of 0.15 and 0.8 respectively.
- The map contains 0 secret sectors, fulfilling the pass criteria.

No technical problems encountered during this test run.

Test Item And Environment

The test evaluated map E1M1 on Freedoom1 using difficulty level 3. The map consists of 8 enemies (Imps, Zombiemen, Shotgun Guy), 1 item (Green Armor), 5 sectors, and 35 linedefs within a 2528x1056 unit dimension. Static complexity analysis indicates a 'hard' difficulty profile due to a high concentration of hitscanners.

The test environment utilized the ViZDoom game engine and the gemini-3.1-flash-lite LLM model via the MCP tool interface. The test was conducted at difficulty level 3 with a 500-tick cap.

Metric	Value	Metric	Value
Status	completed	Duration	22
Actions	3	LLM calls	3
Final HP	100	Kills	1
Position samples	4	Movement units	480.6
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/33675e79-9272-4e02-bedd-d51e89d1449f.mp4		

Static Analysis Context

Metric	Raw WAD	Difficulty 3	Notes
Enemies	8	8	0 hidden by skill/multiplayer flags
Items	1	1	0 hidden by skill/multiplayer flags
Secrets	0	0	Sector special 9
Geometry	35 linedefs	5 sectors	2528 x 1056 units

Selected-difficulty estimate: hard. Health ratio: 0.0. Ammo ratio: 0.0.

Coverage Evaluation

Coverage was limited to the immediate vicinity of the spawn point. While 1 enemy was engaged and 0 secrets were found (none exist in this map), the majority of the 8 spawned enemies were not encountered. The limited duration prevented a meaningful assessment of the entire map layout.

Objectives Covered

- **Engage all enemy types encountered:**

Objectives Omitted

- **Navigate the entire map:** Navigation was incomplete due to a 500-tick timeout.
- **Find secrets where accessible:** Not applicable as the map contains 0 secret sectors.
- **Stress-test geometry, doors, lifts, and switches:** The automated playthrough did not reach areas containing these features.
- **Document resource balance issues:** Incomplete playthrough prevented meaningful resource stress testing.

Uncovered attributes: The automated playthrough could not test map-specific scripting, sector-based triggers, or navigation through map-wide connectivity features due to the limited exploration range.

Defects And Risks

No defects were detected during this test run. The map navigation and combat engagement failed to meet performance thresholds due to the timeout, not due to specific map defects.

Patterns: Insufficient defects to identify patterns.

No defects are currently attached to this run.

Risk Areas

- **Starting sector:** medium High density of hitscanners (62.5% of enemies) combined with low item availability poses a risk to early-run survival.

Activity Log Summary

1) Static analysis phase (pre-tick 17), 2) Map exploration/combat phase (ticks 17-430).

Exploration was significantly limited by the tick cap, preventing movement into the majority of the map layout.

Tick	Event	Position	HP	MCP Action	Stop reason
210	kill	-36.0, -26.7	100	set_objective	objective_set

Limitations And Recommendations

The automated playthrough did not reach the majority of the map sectors, meaning locked doors, switches, and enemy spawns in distant sectors were not encountered.

Good Quality Areas

- **Spawn vicinity:** The automated playthrough was able to successfully identify and neutralize initial threats and attempt to path toward critical items.